Fiscal Year:	FY 2008	**Task Last Updated:**	FY 08/27/2008
PI Name:	Dinges, David F. Ph.D.		
Project Title:	Optical Computer Recognition of Performance Under Stress		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-4209	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	03/01/2004	**End Date:**	04/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	35	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 4/30/2008 per K. Major/NSBRI (3/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Metaxas, Dimitris (Rutgers University) Banks, Siobhan (University of Pennsylvania Health System)		
Grant/Contract No.:	NCC 9-58-NBPF00402		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	(1) Original Aims of Project. This project was focused on the early objective detection of neurobehavioral problems in space flight. The overarching goal was to develop an unobtrusive, automated optical technology to detect psychological distress when astronauts are working during space flight. The primary aim was to determine whether optical computer recognition algorithms based on changes in facial expressions could discriminate behavioral stress induced by low versus high workload performance demands. Astronauts are required to perform mission-critical tasks at a high level of functional capability throughout space flight. There are a number of stressors that can compromise their ability to do so. In order to provide timely countermeasures for stressor-induced impairments in astronauts, objective, unobtrusive measures of the presence of stress reactions—especially during performance demands—are needed. This project was designed to achieve such a measurement methodology, through collaboration between two laboratories: one with expertise in the evaluation of behavioral and physiological responses to stressful performance conditions (D. Dinges, Unit for Experimental Psychiatry, University of Pennsylvania School of Medicine), and the other with expertise in optical computer recognition of human expressions and gestures (D. Metaxas, Computational Biomedicine Imaging and Modeling Center, Rutgers University). The goal of this collaboration was to further develop and test an optically-based computer recognition algorithm of the face to reliably detect the presence of stress. The computer-based recognition system, developed by Prof. Metaxas, utilized automatic optical tracking of human subjects' subtle anatomical and motoric changes in facial expressions. Video input to the system was provided from experiments performed in the laboratory of Prof. Dinges. In this project we experimentally tested the extent to which optical computer recognition algorithms based on facial expressions could detect behavioral stress during cognitive performance. Hypothesis: Optical computer recognition algorithms of the face can reliably discriminate when subjects are undergoing behavioral stress associated with low versus high cognitive workload demands. The project was also intended to serve as an opportunity to improve the technology. Our goal was to deliver an optical computer recognition system capable of identifying stress reactions based on the automated analysis of facial expressions in real-time. (2) Key Findings. In our initial experiments, the validity of the workload paradigm to induce differential levels of stress in facial expressions was established. Basic stress-related facial expressions required to establish a prototypical OCR algorithm to detect such changes were also identified. We established that OCR identification of stress during performance is possible and that the accuracy of the system could be improved. We then further improved the OCR algorithm and the automated application of the deformable masks to video of the moving face to increase practical utility. In more recent experiments, we completed development of an optical algorithm for real-time dynamic tracking of the face using a deformable model-based tracker and Active Shape Modeling. To overcome the limitations of previous optical tracking techniques, Metaxas and colleagues developed a formal framework for the integration of edge detection and optical flow into a deformable model framework and applied it to facial shape and motion estimation. This method used a single camera to track the shape of the face and its movement in 3-dimensional space, and it created a deformable model, incorporating optical flow (an approximation of the motion of objects within a visual representation) into the model as a constraint. Our earlier version of the OCR system, which utilized 3D deformable models for face tracking and Hidden Markov Models (HMM) for stress detection, was 68% accurate in discriminating between high and low stress situations in N=60 subjects. After we reduced computational burden of the OCR system by using Active Shape Models for face tracking and the more efficient Conditional Random Fields for stress detection, accuracy increased to 73%. We have made several other new developments to the OCR system during this period of support: (1) the technique was validated with the use only one camera, where the previous method required two; (2) we improved tracking by using a manifold of faces that helped automatically track the face as the head moves; (3) we added the use of Conditional Random Fields in addition to Hidden Markov Modeling, to the algorithm which improved its computational efficiency; and (4) GABOR filtering (used for edge detection in image analysis) was incorporated into the ASM algorithm to track changes in facial texture, allowing it to identify features (e.g., furrowed brow). (3) Impact of Findings. The study focused on the ability of an unobtrusive, automated optical technology to detect psychological distress (and the need for countermeasures for it) during operational performance. The findings support the hypothesis that machine vision can be used to as an objective, unobtrusive, automated method for the recognition, monitoring, and management of the risks of neurobehavioral dysfunction due to work-related stress in space flight and in many Earth-based safety-sensitive occupations, such as transportation workers (e.g., truck drivers, train conductors, airline pilots); operators in safety-sensitive industries (e.g., power plant control rooms); and military personnel. (4) This project has been completed thus there is no proposed research plan for the coming year.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The study focuses on the ability of an unobtrusive, automated optical technology to detect psychological distress (and the need for countermeasures for it) during operational performance. The knowledge gained has the potential to identify an objective, unobtrusive, automated method for the recognition, monitoring, and management of the risks of neurobehavioral dysfunction due to work-related stress in space flight and in many Earth-based safety-sensitive occupations, such as transportation workers (e.g., truck drivers, train conductors, airline pilots); operators in safety-sensitive industries (e.g., power plant control rooms); and military personnel.
	The overarching goal of the project is to provide an unobtrusive objective detection of stress during performance in space flight using machine vision (computational model-based tracker). We completed development of an optical algorithm for real-time dynamic tracking of the face using a deformable model-based tracker and Active Shape Modeling. To overcome the limitations of previous optical tracking techniques, Metaxas and colleagues developed a formal framework for the integration of edge detection and optical flow into a deformable model framework and applied it to facial shape and motion estimation. This method used a single camera to track the shape of the face and its movement in 3-dimensional space, and it created a deformable model, incorporating optical flow (an approximation of the motion of objects within a visual representation) into the model as a constraint. Our earlier version of the OCR system, which utilized 3D deformable models for face tracking and Hidden Markov

Task Progress:	Models (HMM) for stress detection, was 68% accurate in discriminating between high and low stress situations in N=60 subjects. After we reduced computational burden of the OCR system by using Active Shape Models for face tracking and the more efficient Conditional Random Fields for stress detection, accuracy increased to 73%. The computerized deformable mask of the face has been successfully automated by the Metaxas laboratory to track head and face movements in real time. The cue integration-based tracking system accurately captures rigid and non-rigid parameters of different parts of the face (eyebrows, lips). The labeled sequences are then used to train the recognition system, which consists of generative (Hidden Markov Model) and discriminative (Support Vector Machine) parts that yield results superior to using either approach individually. Accuracy and practical feasibility of the technique is being improved further with automatic multi-resolution selection for the discretization of the mask, and automated face detection and mask initialization algorithms. We have made several other new developments to the OCR system during this period of support: (1) the technique was validated with the use only one camera, where the previous method required two; (2) we improved tracking by using a manifold of faces that helped automatically track the face as the head moves; (3) we added the use of Conditional Random Fields in addition to Hidden Markov Modeling, to the algorithm which improved its computational efficiency; and (4) GABOR filtering (used for edge detection in image analysis) was incorporated into the ASM algorithm to track changes in facial texture, allowing it to identify features (e.g., furrowed brow).
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Articles in Peer-reviewed Journals	Dinges DF, Venkataraman S, McGlinchey EL, Metaxas DN. "Monitoring of facial stress during space flight: optical computer recognition combining discriminative and generative methods." Acta Astronautica. 2007 Feb-Apr;60(4-7):341-50. http://dx.doi.org/10.1016/j.actaastro.2006.09.003 , Mar-2007
Awards	Dinges DF. "1st Place Computational Modeling Prize, “Understanding Decrements in Knowledge Access Resulting from Increased Fatigue“, The Cognitive Science Society, June 2007." Jun-2007
Awards	Dinges DF. "Laurence R. Young Space Biomedical Research Award for contributions to human performance in space, March 2008." Mar-2008
Awards	Dinges DF. "NASA Distinguished Public Service Medal, May 2007." May-2007